

Markscheme

Mathematics: analysis and approaches
Practice question bank

74. (a)

The first, third and ninth terms of an arithmetic sequence, in that order, form a geometric sequence. The first term of the arithmetic sequence is 4 and the common difference is $d \neq 0$.

Find the value of d .

write the three arithmetic terms: 4, $4 + 2d$, $4 + 8d$ (M1)

geometric condition: $(4 + 2d)^2 = 4(4 + 8d)$ (M1) A1

$$4d^2 - 16d = 0 \Rightarrow 4d(d - 4) = 0 \quad \text{A1}$$

reject $d = 0$; $d = 4$ R1

[5 marks]

(b)

Hence find the common ratio of the geometric sequence formed.

the terms are 4, 12, 36 (M1)

$$r = 3 \quad \text{A1}$$

[2 marks]